VPC PEERING

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Batch : AWS Cloud Computing

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APPLICATION : INTERNSHIP_17893631566aa783d4b0c90

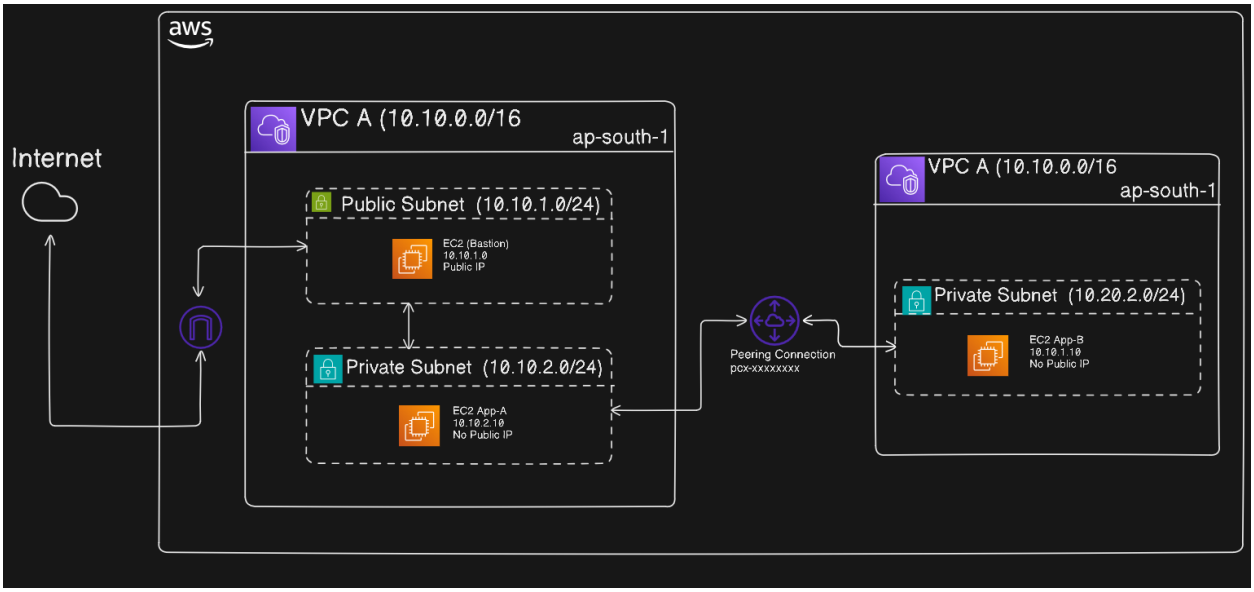
1. Project Overview

The objective of this project is to establish private IPv4 connectivity between two Amazon Virtual Private Clouds
(VPCs) using a VPC peering connection. The design keeps both application servers private and uses one public
bastion host in VPC A for administrative SSH access. Connectivity depends on the peering connection, correct routes
in both directions, and security-group rules that allow only the required traffic

Project objectives

- Plan non-overlapping VPC and subnet CIDR ranges.
- Create public and private subnets with explicit route-table associations.
- Create and activate VPC peering between VPC A and VPC B.
- Configure forward and return routes through the peering connection.
- Restrict inbound SSH and ICMP traffic with security groups.
- Validate private connectivity and document controlled failure and recovery.

2. Architecture



Design notes

VPC A uses 10.10.0.0/16 and VPC B uses 10.20.0.0/16; these CIDRs do not overlap. VPC A contains the public bastion and App-A. VPC B contains App-B and has no Internet Gateway. VPC peering provides private routing between the VPCs, but it does not make VPC B transit through VPC A's Internet Gateway.

3. Resource Inventory

The following resources are planned for the AWS VPC Peering project in the Asia Pacific (Mumbai) region (ap-south-1).

3.1 VPC and Subnet Details

Resource	Name	IPv4 CIDR
VPC A	ewb-vpc-a	10.10.0.0/16
VPC B	ewb-vpc-b	10.20.0.0/16
Public Subnet A	ewb-a-public	10.10.1.0/24
Private Subnet A	ewb-a-private	10.10.2.0/24
Private Subnet B	ewb-b-private	10.20.1.0/24

3.2 Internet Gateway and Route Tables

Resource	Name	Configuration
Internet Gateway	ewb-igw-a	Attached to VPC A
Public Route Table	ewb-rt-a-public	Associated with ewb-a-public
Private Route Table A	ewb-rt-a-private	Associated with ewb-a-private
Private Route Table B	ewb-rt-b-private	Associated with ewb-b-private

3.3 EC2 Instance Details

Instance	Name	Subnet	Private IPv4 Address	Public IPv4 Address
Bastion Host	ewb-bastion	ewb-a-public	10.10.1.10	Record actual address
Application Server A	ewb-app-a	ewb-a-private	10.10.2.10	None
Application Server B	ewb-app-b	ewb-b-private	10.20.1.10	None

Operating System: Amazon Linux 2023

EC2 Key Pair: ewb-lab

Instance Type: t3.micro, if used and permitted by the lab account.

3.4 Security Groups

Security Group	Associated VPC	Purpose
ewb-sg-bastion	ewb-vpc-a	Allows SSH access from the student's public IPv4 address
ewb-sg-app-a	ewb-vpc-a	Allows SSH from the bastion and ICMP Echo Requests from App-B
ewb-sg-app-b	ewb-vpc-b	Allows SSH from the bastion and ICMP Echo Requests from App-A

3.5 VPC Peering Connection

Field	Details
Connection Name	ewb-a-b-peer
Requester VPC	ewb-vpc-a
Accepter VPC	ewb-vpc-b
AWS Region	ap-south-1
Peering Connection ID	Record actual pcx-... ID
Connection Status	Record the actual status

3.6 Addressing and Routing Summary

VPC A uses the CIDR range 10.10.0.0/16, while VPC B uses 10.20.0.0/16. These ranges do not overlap, allowing the planned VPC peering connection.

The public route table in VPC A contains a default route (0.0.0.0/0) to the Internet Gateway and a route to VPC B through the peering connection. VPC A's private route table and VPC B's private route table contain routes to the opposite VPC's CIDR range through the same peering connection.

The application servers remain private and do not receive public IPv4 addresses.

Note: This inventory lists the planned project configuration. Actual resource IDs, public IP addresses, connection status, and implementation details must be verified against the AWS environment before submission.